

Instructor Key: Day 8 Mastery Quiz: Debugging and Test Design

Expected-versus-actual route to minimal fix to regression test

Instructor key. Same layout as student quiz; solutions filled in response boxes. Print duplex on one sheet.

Name		Section		Date	
Score	_____/10		Mastery check		secure / developing / needs support

Engineering situation and quiz rule

A checkout script passed a quick glance but failed a boundary case. Use expected-versus-actual evidence to locate the bug, propose a minimal fix, and name a regression test.

1 Reference code for Items 1–10

[trace](#) [explain](#)

```
1 charge_percent = 78
2 calibration_label = "current"
3 damage_note = ""
4
5 if damage_note != "":
6     route = "repair bench"
7 elif charge_percent > 80:
8     route = "charge station"
9 elif calibration_label != "current":
10    route = "calibration bench"
11 else:
12    route = "release kit"
13
14 print(route)
```

2 Items 1–5: trace the route evidence

[predict](#) [route](#)

Pt.	Prompt	My response
1	For charge_percent 78, what route should the checkout process choose?	Key: "charge station" because 78 is below the minimum charge threshold.
1	What route does the buggy code actually print?	Key: "release kit" because damage is false, charge_percent > 80 is false, calibration is current, so else runs.
1	Which condition contains the likely bug?	Key: charge_percent > 80 is the wrong direction for the charge-station hold.
1	Write the minimal corrected condition for the charge station route.	Key: Use charge_percent < 80 for the charge station route.
1	Why is changing the final else branch not the best minimal fix?	Key: The else should mean no hold conditions are true; changing it would hide the real faulty charge condition.

Pt.	Prompt	My response
1	Name one boundary test for the charge rule.	Key: Test 79, 80, and/or 81. A secure answer states expected route for the chosen boundary.
1	Name one regression test that should still release a clean kit.	Key: Example: damage blank, charge_percent 82, calibration current, initials present should release or continue as clean according to the route map.
1	If charge_percent is 80, should the kit charge or continue? Explain using the intended rule.	Key: At 80 the kit should not go to charge station if the intended hold is below 80.
1	Write one expected-versus-actual sentence for the original bug.	Key: Expected charge station for 78; actual release kit; likely cause is the reversed > comparison.
1	What support route would help someone who sees the wrong output but cannot locate the cause?	Key: Build an expected/actual/cause/fix/test row before editing code.

Support route, not scored

Separate expected route from actual output before editing. The fix should target the smallest condition that explains the mismatch.